POWER BI · LESSON 5 OF 8

Visuals, Filters, Drill-through, Tooltips, Buttons & Report Design

Turn a model into an interactive, well-designed report. Choose visuals, format them (cards, matrix, charts, gauge, KPI, maps, images), use visual, page and report filters, edit interactions, drill-down and drill-through, custom tooltips, bookmarks and buttons, conditional formatting, themes and design principles.

Power BI classes 2–4 | Power BI Desktop 2026 | Intermediate | 150 min

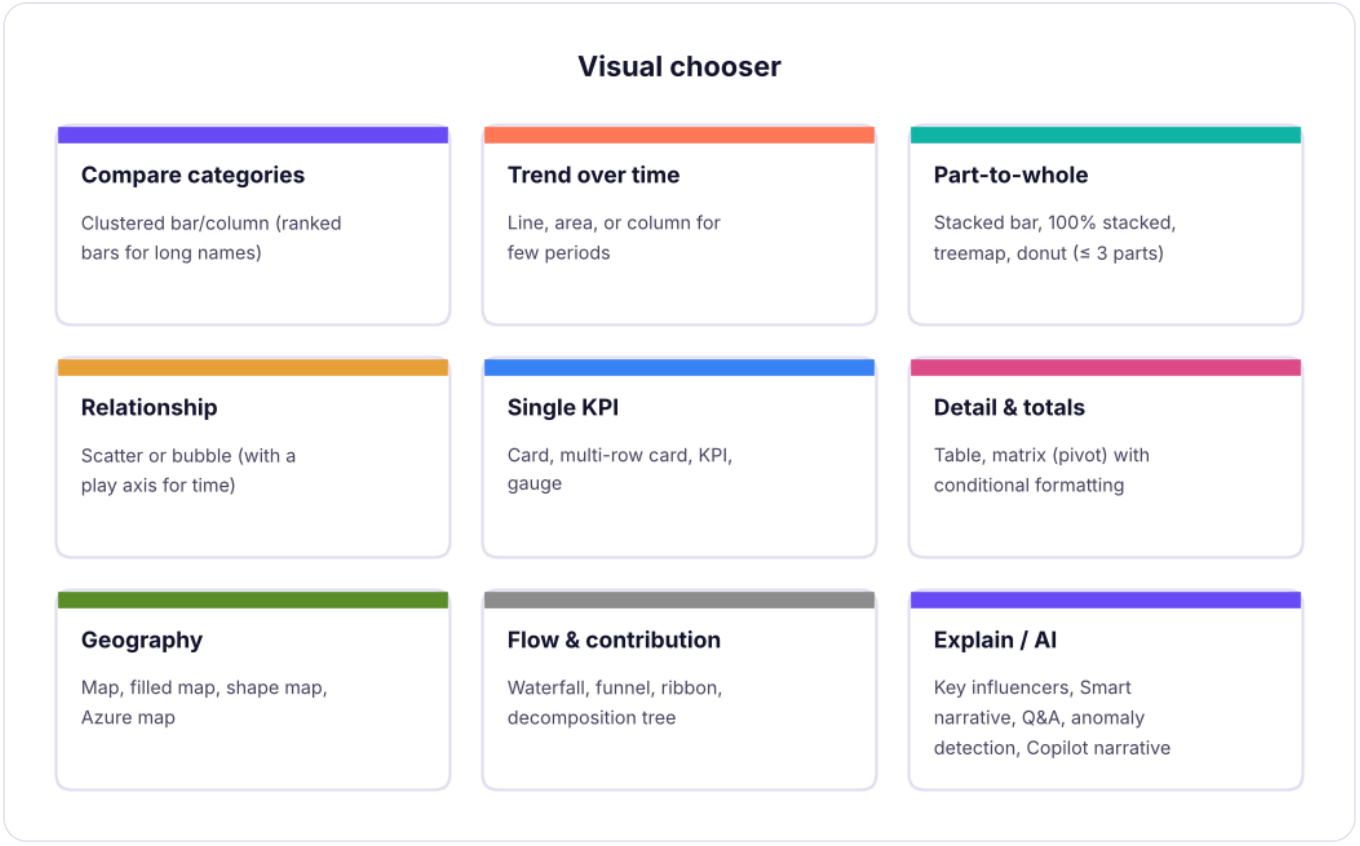
By the end of this lesson you can

- Pick the right visual for each question and configure its field wells and formatting
- Apply visual, page and report-level filters, Top N, relative date filters and slicer sync
- Control cross-filtering with Edit interactions; add drill-down, drill-through and report-page tooltips
- Build navigation with buttons, bookmarks, page navigation and a selection pane
- Apply themes and conditional formatting, and follow layout, colour and accessibility rules

Datasets used

`adventure_cycles/fact_sales.csv` `adventure_cycles/dim_product.csv`
`adventure_cycles/dim_territory.csv` `adventure_cycles/dim_customer.csv`

Choosing the visual



Match the visual to the question.

Visual	Fields	Good for	Watch out
Card / multi-row card	Values	One or few headline numbers	Format display units (K, M, Cr) and add a clear label
Clustered bar / column	Axis, Values, Legend	Comparisons	Sort descending, start at zero, ≤ 15 bars
Stacked / 100% stacked	Axis, Legend, Values	Composition	Only the bottom segment is easy to compare
Line / area	X-axis (date), Y-axis, Legend	Trends	Limit to ~5 lines; use a continuous date axis
Line + clustered column	Shared axis, column values, line values	Two measures with different scales (secondary axis)	Label both axes
Waterfall	Category, Y-axis	Bridge from start to end (profit walk)	Needs an ordered category
Scatter / bubble	X, Y, Size, Details, Play axis	Relationships and outliers	Add Details so each point is an entity
Treemap	Group, Values	Hierarchical share of total	Many tiny rectangles are unreadable
Donut / pie	Legend, Values	2–3 parts	More than 4 slices = use a bar
Matrix	Rows, Columns, Values	Pivot tables with drill and subtotals	Use conditional formatting for heat maps
Table	Columns	Exact values, detail list	Keep it short; a table is not a dashboard
KPI / **Gauge**	Value, Target, Trend axis	Progress to a target	Gauges waste space; prefer KPI cards
Funnel	Stage, Values	Drop-off between stages	
Map / Filled map	Location, Size, Legend	Geography	Geocoding needs clean place names (`City, State, Country`)
Slicer	Field	Filter the page	Styles: list, dropdown, tiles, between (range), relative date
Decomposition tree	Analyze, Explain by	Drill into drivers	Interactive exploration

Visual	Fields	Good for	Watch out
Key influencers	Analyze, Explain by	AI finds what drives a metric	Needs a binary/numeric target
Smart narrative	Auto-text	Written summary that updates	Check the wording

Formatting essentials

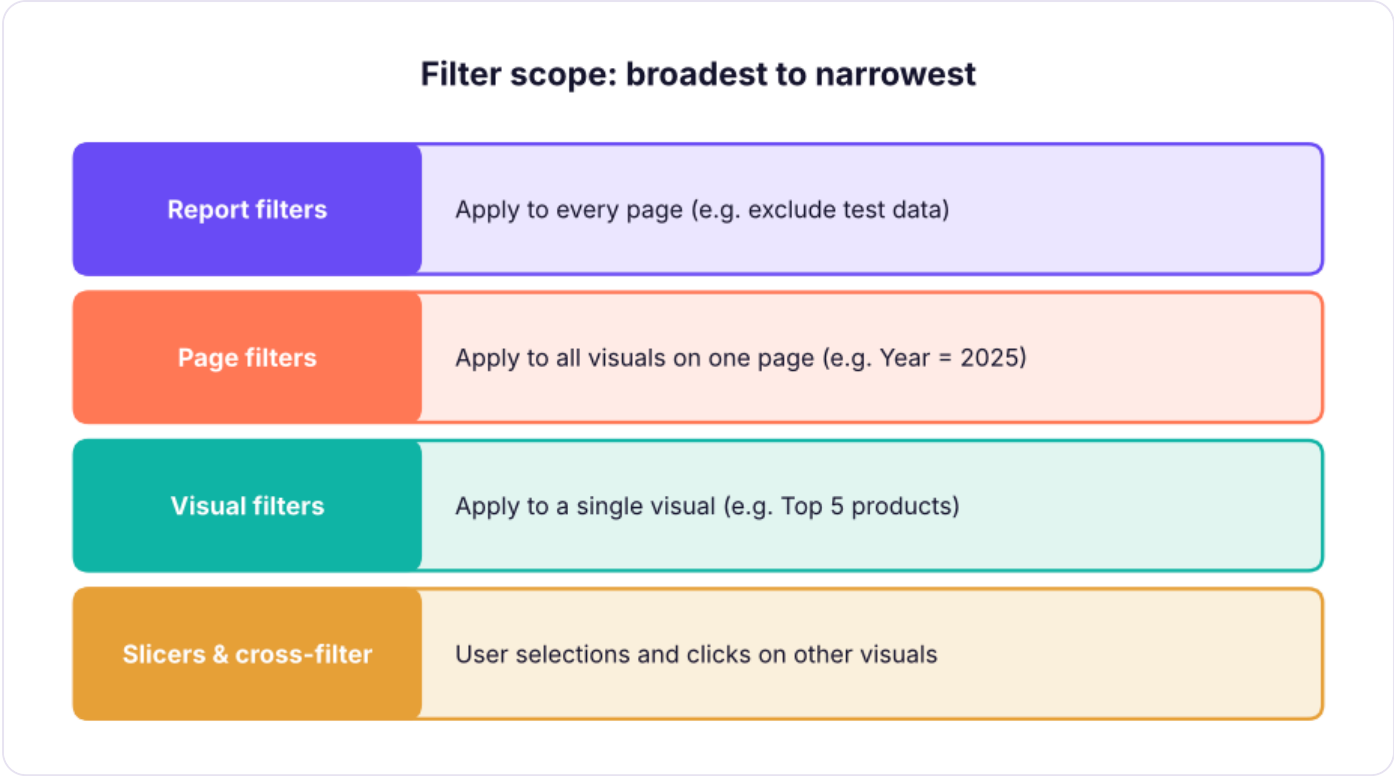
Select a visual → **Format visual** (paint-roller icon): **Visual** tab (axis, labels, colours, legend, gridlines, title) and **General** (title, size, position, background, border, alt text).

Task	Where
Display units and decimals	Format → Callout value / Data labels → Display units: Thousands, Lakhs (not built in; use Millions) or custom format strings
Sort	`…` menu → Sort axis (by value or by another field)
Colour a highlighted series	Format → Colors → per-category colours or conditional
Titles that state the insight	General → Title → fx (dynamic title from a measure)
Accessibility	General → Alt text; keep contrast high; do not rely on colour alone
Copy formatting	Home → **Format painter**
Align / distribute	Format → Arrange → Align and distribute; hold `Shift` to multi-select
Group visuals	Select several → Format → Group

TIP **Dynamic titles**

Create a measure `Title Sales = "Sales by " & SELECTEDVALUE ( dim_territory[country], "all countries" ) & " in " & SELECTEDVALUE ( dim_date[year], "all years" )` and bind it in Title → fx → *Field value*. The title now reflects the user's selections.

Filters: three levels



The three filter levels. Visual filters affect one visual, page filters one page, report filters every page.

Use the **Filters pane** (right side). Filter types:

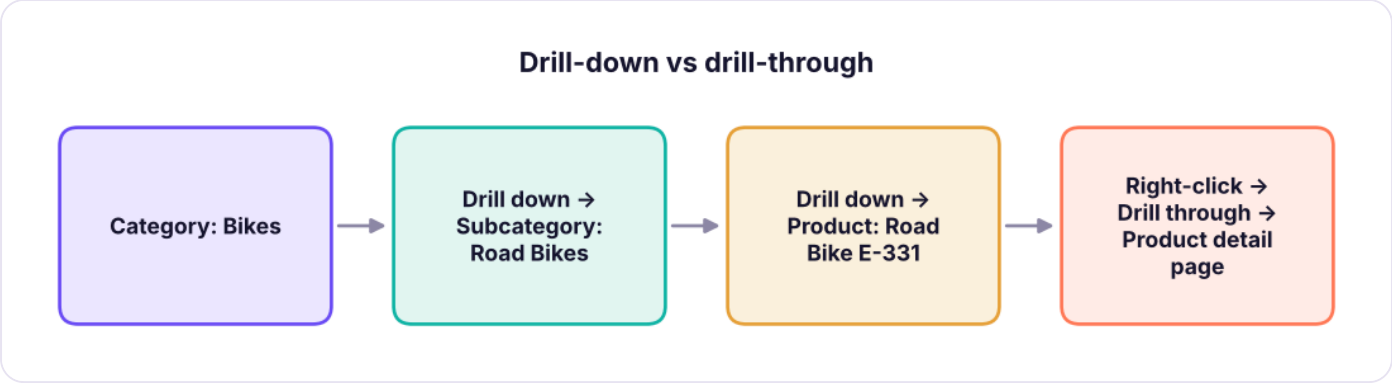
Filter type	Behaviour
Basic	Tick values from a list
Advanced	"contains", "starts with", "is greater than", with AND/OR
Top N	Top/Bottom N items by a measure (even one not in the visual)
Relative date	"Last 6 months", "this year", "next 7 days"
Relative time	Last 4 hours etc.

Tips: **Require single selection** in a slicer; **Sync slicers** across pages (View → Sync slicers); the **"Include" / "Exclude"** options on a selection filter in or out; lock the filters pane or hide it for consumers.

Edit interactions

By default clicking a bar **cross-filters** other visuals. Select the source visual → Format → **Edit interactions** → then on each target visual click the **Filter**, **Highlight** or **None** icon. Example: a KPI card for "overall sales" should ignore the clicks on the country chart (set to None), while the product chart should filter.

Drill-down and drill-through



Drill-down moves through a hierarchy inside one visual. Drill-through jumps to another page that is filtered by the selected item.

- **Drill-down**: needs a hierarchy (Lesson 2). Use the arrow buttons at the top of the visual: ↑ drill up, ↓ drill mode, ⇊ go to the next level in the hierarchy, ⇈ expand all.
- **Drill-through**: create a **detail page** → in the Visualizations pane, drag `product` into **Drill through → Add drill-through fields here** → Power BI adds a back button automatically. Users right-click a product in any visual → **Drill through → Product detail**. Use it for "customer 360", "order detail" and "product detail" pages. **Keep all filters** keeps slicer context.
- **Cross-report drill-through** links to a separate report in the Service.

Report-page tooltips

A **tooltip page** shows a mini-report when hovering over a data point.

STEP BY STEP **Build one**

1. Add a new page; Format page → Page information → **Allow use as tooltip** on; Page size → **Tooltip**.
2. Add visuals (a KPI, a small trend line, top 3 products) using fields that respond to the hovered item.
3. In the main visual: Format → General → **Tooltips** → Type **Report page** → choose the page.
4. Hide the tooltip page from navigation (right-click the tab → Hide page).

Buttons, bookmarks and navigation

Feature	What it does	Use
Buttons (Insert → Buttons)	Back, Bookmark, Page navigation, Q&A, Web URL, Apply all slicers, Clear all slicers	Menus, navigation bars, reset filter
Bookmarks (View → Bookmarks)	Save the current state (filters, slicer selections, visibility, spotlight) of a page	Story-telling views, toggle between visuals
Selection pane (View → Selection)	List, name, show or hide every object on a page	Required for bookmark toggles
Page navigator (Insert → Navigator)	Auto-generated tab buttons for pages or bookmarks	Menu
Conditional visibility	Bookmarks that show/hide visuals	"Chart ⇄ Table" toggle

STEP BY STEP

Chart / table toggle with bookmarks

1. Place a column chart and a table on top of each other.
2. Selection pane: hide the table → View → Bookmarks → **Add** → name it "Chart view"; make the table visible and hide the chart → Add → "Table view". Untick **Data** in each bookmark's options so slicers are not reset.
3. Insert two buttons and set Action → **Bookmark** → choose the matching bookmark.

Conditional formatting

Where	How
Table / matrix cells	Format → Cell elements → Background colour / Font colour / **Data bars** / **Icons** / Web URL → fx → *Gradient*, *Rules* or *Field value*
Column / bar colour	Format → Colors → fx → rules (green above target, red below)
Card text colour or title	fx on the callout/title
Based on a measure	Create a measure returning a hex colour: `Margin Colour = IF ( [Profit Margin %] ≥ 0.35, "#1E8E5A", "#C0392B" )` then Field value

Subcategory	Sales (₹ M)	Profit (₹ M)	Margin %
Mountain Bikes	161.9	54.9	33.9%
Road Bikes	155.7	56.1	36.0%
Touring Bikes	149.2	55.2	37.0%
Frames	76.1	27.3	35.8%
Wheels	33.4	10.5	31.3%
Helmets	8.2	2.7	33.1%
Jerseys	5.8	2.2	37.2%
Gloves	2.7	0.8	30.8%
Bottles	1.6	0.5	34.0%

A matrix with conditional formatting: a heat-map background on sales and traffic-light colours on margin (rules: ≥ 36% green, otherwise amber/red).

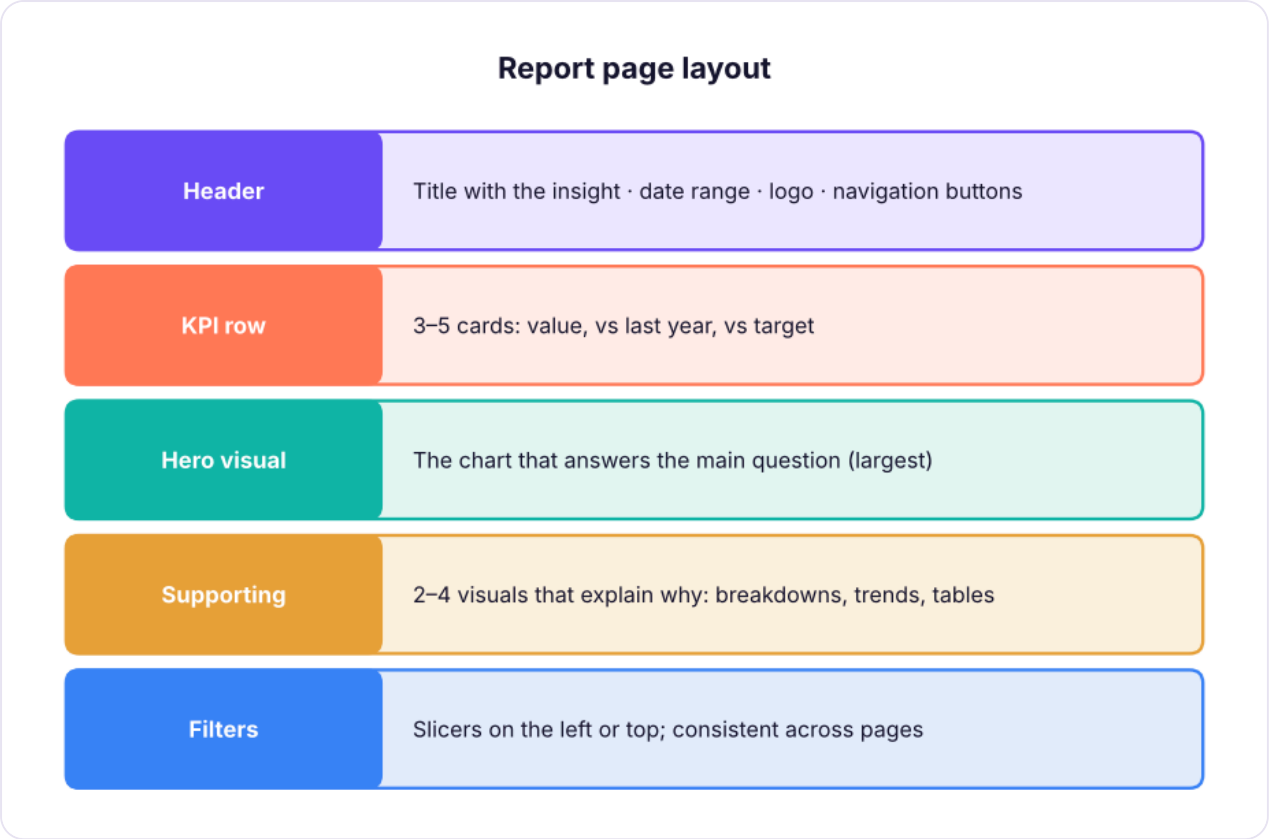
Themes

View → Themes offers built-in themes; you can download community themes or **Customize current theme** (colours, fonts, defaults). A **theme JSON file** (View → Themes → Browse for themes) applies one consistent look to every visual:

```json
{
  "name": "SkillNavora",
  "dataColors": ["#6C4CF5", "#FF7A59", "#12B5A6", "#E9A23B", "#3B82F6", "#E04F8A"],
  "background": "#FFFFFF", "foreground": "#17142E", "tableAccent": "#6C4CF5",
  "textClasses": { "title": { "fontFace": "Segoe UI Semibold", "fontSize": 14, "color":
"#17142E" } }
}
```

Apply the company theme to every report so they look consistent, and edit colours in one place.

Report design principles



A layout that works: headline KPIs on top, the main insight chart large, supporting visuals below, filters on one side, consistent spacing.

1. **Start from the audience and decision**, not from the data: what will they do differently?
2. **Hierarchy**: most important top-left; big numbers first; details lower.
3. **Less is more**: 5–8 visuals per page; remove gridlines, borders and chart junk.
4. **Consistent colour logic**: one brand colour plus one highlight; red and green only for bad/good; the same colour always means the same thing.
5. **Align and space** visuals on a grid; same sizes for similar charts.
6. **Label for meaning**: insight titles, units (₹, %), clear axis titles, "as of" dates.
7. **Accessible**: contrast, alt text, tab order (Format → Tab order), do not use colour alone, test on mobile layout (View → Mobile layout).
8. **Performance**: fewer visuals, summarised data, no huge tables; check Optimize → Performance analyzer.

Q&A, images and mobile

- **Q&A visual**: users type "total sales by country in 2025" and get a chart. Improve it with **synonyms** (Model view → Linguistic schema / Q&A setup).

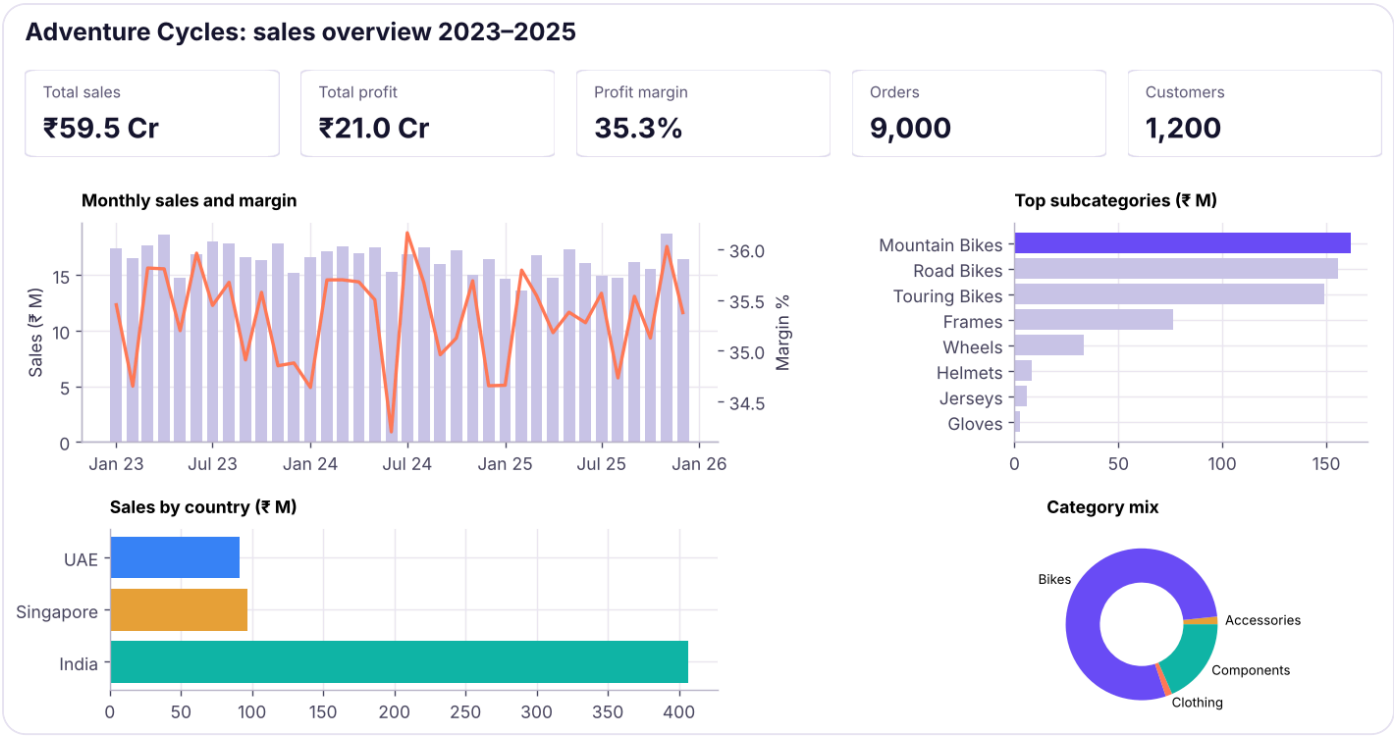
- **Images and logos**: Insert → Image (or text box). For images **in data** (product photos), set the column's **Data category → Image URL**.
- **Mobile layout**: View → Mobile layout; drag key visuals into a phone-shaped canvas.
- **Accessibility and exporting**: export to PDF or PowerPoint (File → Export), analyse in Excel, subscribe in the Service.

> **2026 UPDATE** **Copilot and AI visuals in 2026**
>
> Copilot can generate a report page from a prompt, suggest visuals for your fields, write summaries ("Summarise this page") and answer questions about the data. **Key influencers**, **Decomposition tree** and **Smart narrative** are built-in AI visuals; **anomaly detection** (Analytics pane) flags unusual spikes. Always verify AI-generated visuals and text against real numbers.

A mini case: the Adventure Cycles overview page

Plan the page before you build it:

Element	Fields / measure	Visual
Title	Dynamic title measure	Text box
KPI row	`Total Sales`, `Total Profit`, `Profit Margin %`, `Orders`, each with vs-last-year	4–5 cards
Hero	Monthly sales and profit margin	Line + column
Breakdown 1	Sales by category (Top N subcategories)	Bar
Breakdown 2	Sales by country	Map or bar
Drill-through	Product detail page	Table + KPIs
Filters	Year, Country, Category	Slicers



The Adventure Cycles overview page from the real data: KPI cards, monthly trend, subcategory ranking and country share.

Summary

KEY TAKEAWAYS

- Choose visuals from the question; format with insight titles, units, sorting and accessible colours.
- Filters work at **visual, page and report** level (Basic, Advanced, Top N, Relative date); use slicers, sync them across pages, and **Edit interactions** to control cross-filtering.
- **Drill-down** needs a hierarchy; **drill-through** jumps to a detail page; **tooltip pages** show mini-reports on hover.
- **Buttons + bookmarks + the Selection pane** build navigation and chart/table toggles.
- Use **conditional formatting** (rules, gradients, field value from a measure) and **themes** for consistency.
- Design for the decision: hierarchy, few visuals, consistent colour logic, alignment, labels, accessibility and mobile layout.

Quick quiz

6 questions · tap an answer to check it

1. Which filter level affects every page of a report?

A Visual-level

B Page-level

C Report-level (All pages)

D Slicer-level

Answer: C. Report filters apply to all pages.

2. How do you jump from a product in a chart to a detail page for that product?

A Drill-down

B Drill-through (create a drill-through page with the product field)

C Add a slicer

D Use a tooltip

Answer: B. Drill-through passes the selected value as a filter to another page.

3. What do bookmarks capture?

A Only the page name

B The state of a page: filters, slicers, visibility and more

C The data model

D The theme

Answer: B. Bookmarks save a view that buttons can switch to.

4. Which feature stops a card from reacting when you click a bar chart?

A Sync slicers

B Edit interactions → set the card to None

C Drill-through

D Alt text

Answer: B. Edit interactions controls how visuals filter each other.

5. When is a donut chart appropriate?

A With 12 categories

B With 2–3 parts of a whole

C To show a time trend

D To compare 20 products

Answer: B. More than a few slices are hard to compare; use bars.

6. What is the benefit of a theme JSON file?

A One consistent set of colours and fonts applied to all visuals and reusable across reports

B It refreshes data

C It creates relationships

D It adds measures

Answer: A. Themes standardise the look and make branding quick.

EXERCISE

Build the overview page

Using the Adventure Cycles model: (1) build the overview page above; (2) add a **Year** slicer and a **Country** slicer; (3) create a drill-through page for **Product** with sales, profit, margin and a monthly trend; (4) add a bookmarked toggle between "Sales" and "Profit" using a field parameter or bookmarks; (5) apply conditional formatting to the margin in a matrix and write a dynamic title.

Show solution (worked answer)

(1) Cards for the five measures; line + clustered column for monthly sales and margin; bar for top subcategories; bar for country; donut for category mix. (2) Slicer visuals with `dim_date[year]` (tile style) and `dim_territory[country]` (dropdown). (3) New page → drag `dim_product[product]` into Drill through; add a back button. (4) Modeling → New parameter → Fields → add `Total Sales`, `Total Profit`; use the parameter on the chart values; place its slicer as a toggle. (5) Matrix of `subcategory` × `country` with `Profit Margin %`; Cell elements → Background colour → Rules; title measure with `SELECTEDVALUE`.

INTERVIEW CORNER

Questions you may be asked

Q: How do you design an effective Power BI dashboard? Start from the audience and decision; put headline KPIs top-left; use few visuals with insight titles; consistent colour and spacing; allow drill-down/through; test with users; check performance and accessibility.

Q: Drill-down vs drill-through? Drill-down navigates levels of a hierarchy within a visual; drill-through navigates to another page filtered by the selected item.

Q: What are the filter levels and how do slicers differ? Visual, page and report filters are set by the author in the Filters pane; slicers are on-canvas controls that users change, and they can be synced across pages.

Q: How would you let users toggle between a chart and a table? Two bookmarks that show/hide the visuals through the Selection pane, and two buttons linked to those bookmarks.

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